Part B : Criteria Summary

1 OUTCOME-BASED CURRICULUM (120)

1.1 Vision, Mission and Program Educational Objectives (PEOs) (35)

1.1.1 State the Vision and Mission of the Institute and the Department (5)

Vision of the institute	To become a globally recognized academic institution, striving for excellence in education, research, and technological service while aligning with social, economic, and ecological environments
Mission of the institute	To educate students at Undergraduate, Post Graduate, Doctoral and Post Doctoral levels to perform challenging engineering and managerial jobs in industry To provide excellent research and development facilities to take up Ph.D programmes and research projects To develop effective teaching and learning skills and state-of-the-art research potential of the faculty To build national capabilities in technology, education and research in emerging areas To provide excellent technological services to satisfy the requirements of the industry and overall academic needs of society
Vision of the Department	The Mechanical Engineering Department of Birla Institute of Technology, Mesra, Ranchi strives to be globally recognised for quality engineering education and research leading to well qualified engineers, academicians and researchers who are innovative, entrepreneurial and successful in achieving excellence in their field of study.
Mission of the Department	

Mission No.	Mission Statements
M1	To impart quality education to the students and enhancing their knowledge and skills to be globally competitive Mechanical Engineers.
M2	To maintain state of the art research facilities to provide its students and faculty to create, interpret, apply, and disseminate knowledge with an understanding of the limitations.
M3	To develop linkages and interaction with industry, R & D organization, and educational institution for excellence in consultancy practices, research, and teaching.
M4	To provide conducive environment for learning, creativity and problem-solving skill.

1.1.2 State PEOs of the Program (5)

PEO No.	Program Educational Objectives Statements
PEO1	To provide a quality undergraduate education for students entering the mechanical engineering profession of seeking carriers in related fields.
PEO2	To advance scientific knowledge through basic and applied research through scholarly publication, conferences, and continuing education
PEO3	To enable to acquire knowledge of relevant technologies and multidisciplinary fields including broad social, ethical, and environmental issues within which the engineering is practiced.
PEO4	To develop problem solving approach using analytical abilities, effective communication skills and teamwork.
PEO5	To create awareness and understanding related to social issues, apart from developing a sense of commitment to the community and profession with sincere involvement.

1.1.3 Process of Defining Vision, Mission and PEOs (10)

The Department of Mechanical Engineering followed a systematic, inclusive, and consultative process for defining its Vision, Mission, and Programme Educational Objectives (PEOs). The process involved active participation from all internal and external stakeholders and was carried out in the following steps:

STEP I:

Several departmental meetings and brainstorming sessions were conducted with the active participation of all faculty members of the Department of Mechanical Engineering. During these deliberations, due consideration was given to the Institute's Vision and Mission as well as the guidelines prescribed by regulatory and accreditation bodies such as the National Board of Accreditation (NBA) and NAAC. After extens discussions, a draft version of the Department Vision, Mission, and Programme Educational Objectives (PEOs) was prepared.

STEP II:

Feedback on the draft Vision, Mission, and PEOs was obtained from internal stakeholders, including:

- The Management of the Institute (Vice Chancellor, Dean – Academic Programmes, Dean – Faculty Affairs & Sponsored Research)
- All Faculty Members
- Support Staff
- Current Students

STEP III:

Feedback was also collected from external stakeholders, comprising:

- Alumni
- Employers and Industry Representatives
- Parents

STEP IV:

All feedback received from internal and external stakeholders was carefully analyzed and reviewed. Based on the inputs, a revised second draft of the Vision, Mission, and PEOs was prepared. This draft was fur discussed and finalized by the Departmental Academic Committee (DAC). To ensure continuous improvement, industry relevance, and better alignment with current industrial standards, several new courses we introduced in the curriculum.

STEP V:

After final ratification by the Departmental Academic Committee (DAC), the revised Vision, Mission, and PEOs were placed before the Board of Studies (BoS) for approval. The BoS thoroughly reviewed the documents and formally accepted the same.

STEP VI:

The Vision, Mission, and PEOs approved by the Board of Studies were presented and discussed in the Academic Council meeting of the Institute. Upon approval by the Academic Council, the final Vision, Mission and PEOs of the Department of Mechanical Engineering were officially implemented and displayed on the institute website and other prominent locations.

Flowcharts (Figures 1.1.3.1 and 1.1.3.2) illustrating the complete process followed for defining the Vision, Mission, and PEOs of the Department of Mechanical Engineering are provided below:

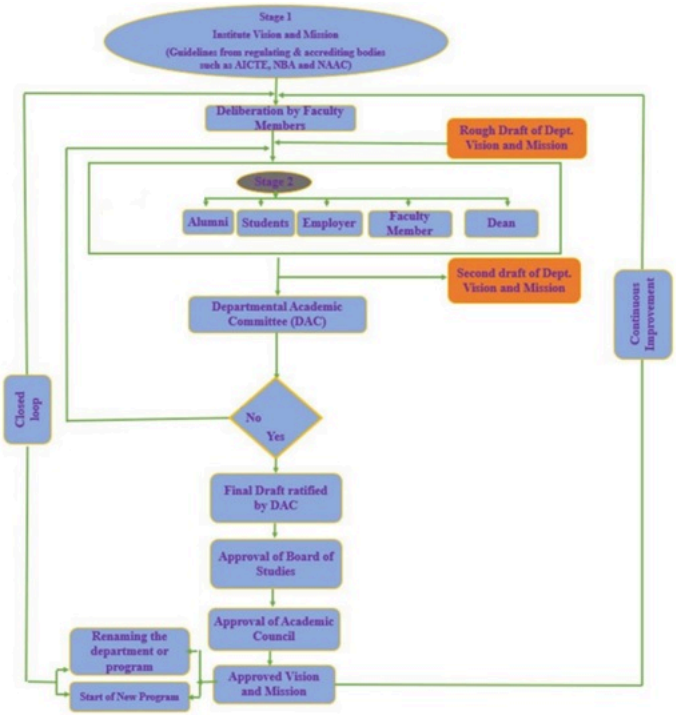


Figure 1.1.3.1: Flowchart showing the process followed for establishing the Vision and Mission.

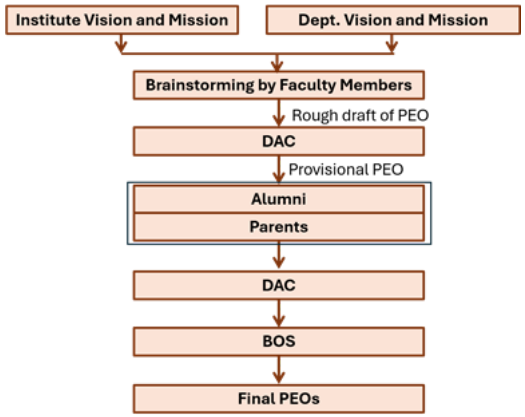


Figure 1.1.3.2: Flowchart depicting the process followed for establishing the PEOs.

1.1.4 Dissemination of Vision, Mission and PEOs (5)

The Vision, Mission, and PEOs are published in the Website of the University as well as in the Laboratories of the Department. Stakeholders are made aware of the Vision, Mission, and PEOs.

INTERNAL STAKE HOLDERS:

1. Management of the Institutes including Vice Chancellor, Dean Academic Programme, Dean Faculty Affairs and Sponsored Research
2. All Faculty
3. Support Staff
4. Students

EXTERNAL STAKE HOLDERS:

1. Alumni
2. Employers
3. Industries
4. Experts from other reputed institutions Posters, and Flyers are printed and disseminated as given in the following Table:

Medium/Location of publication	**INSTITUTE**		**Department**		
	Vision	Mission	Vision	Mission	PEO
Institute Website	Y	Y	Y	Y	Y
Department Notice Board	Y	Y	Y	Y	Y
Common areas- Corridors	Y	Y	Y	Y	Y
Staff rooms, labs, classrooms	Y	Y	Y	Y	Y

Following figures (figs. 1.1.4.1, 1.1.4.2, 1.1.4.3, 1.1.4.4, 1.1.4.5, 1.1.4.6) exhibit different locations where the CO, PO and PEOs are disseminated.

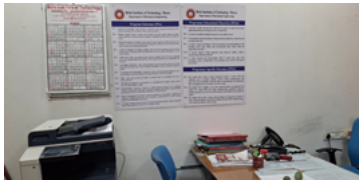



Figure 1.1.4.1: Display at HOD cabin and office

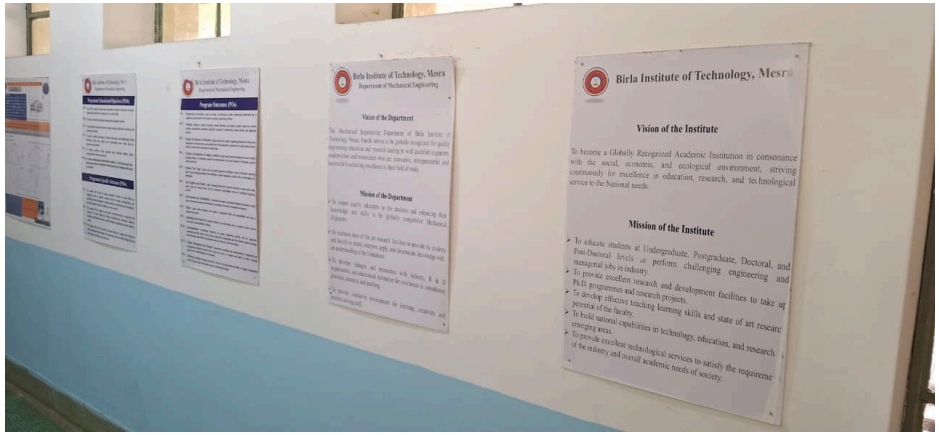


Figure 1.1.4.2: Display at Mechanical Wing



Figure 1.1.4.3: Display at IC Engine laboratory



Figure 1.1.4.4: Display at DOM laboratory

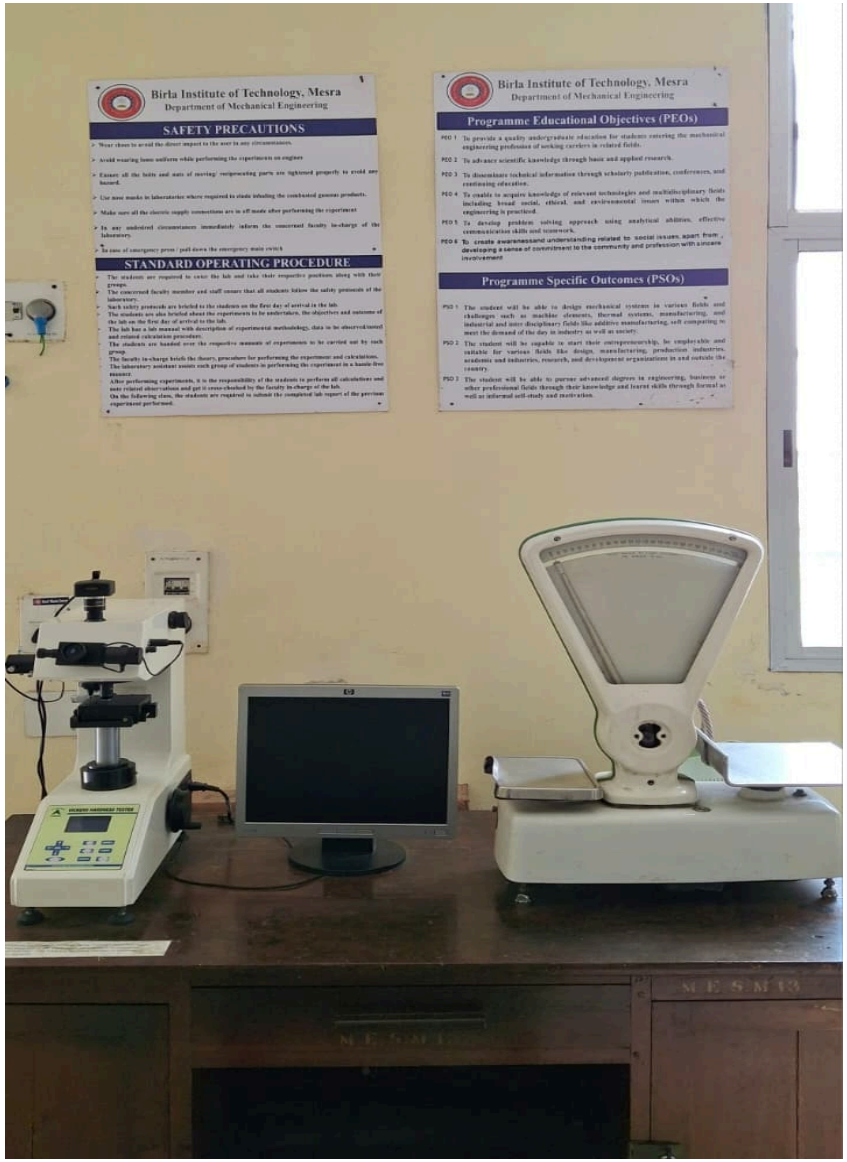


Figure 1.1.4.5: Display at SOM laboratory

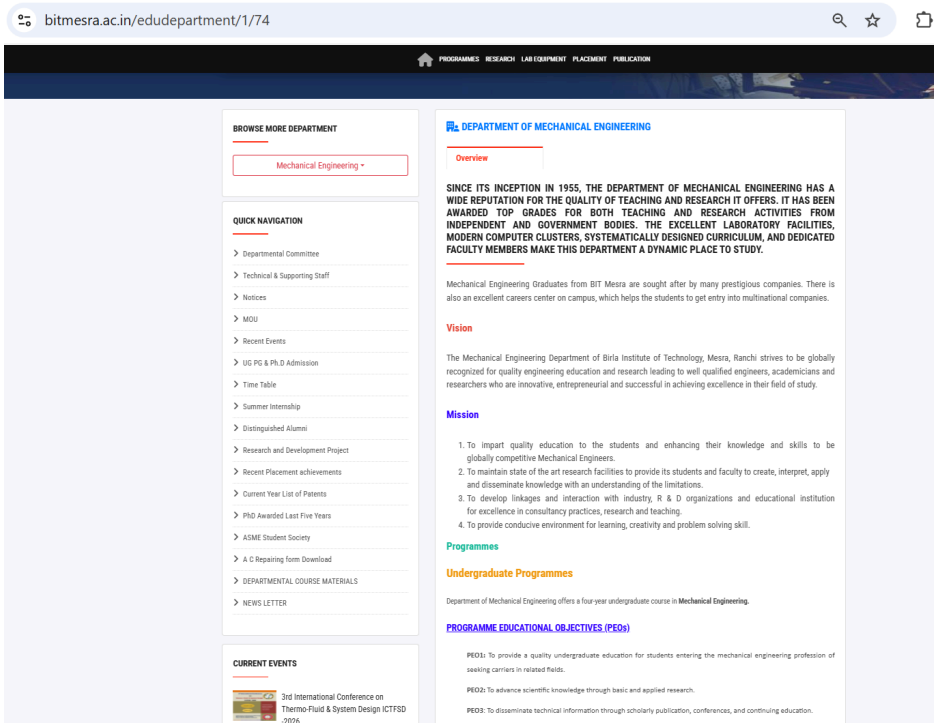


Figure 1.1.4.6: Display at Institute website department page

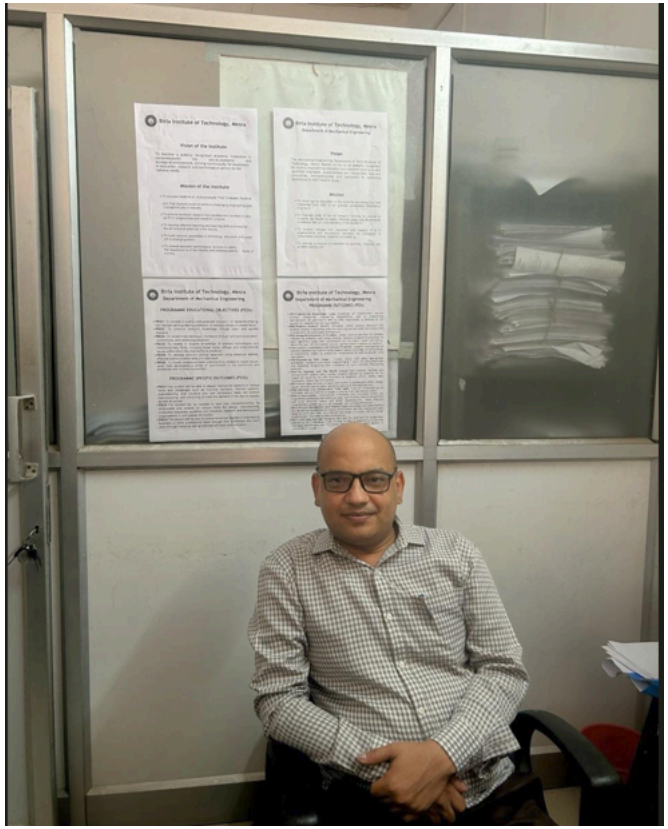


Figure 1.1.4.7: Display at staff room

1.1.5 Mapping of PEOs with Mission (10)

PEO Statements	M1	M2	M3	M4
To provide a quality undergraduate education for students entering the mechanical engineering profession of seeking carriers in related fields.	3	2	2	3
To advance scientific knowledge through basic and applied research through scholarly publication, conferences, and continuing education	2	3	3	2
To enable to acquire knowledge of relevant technologies and multidisciplinary fields including broad social, ethical, and environmental issues within which the engineering is practiced.	2	3	3	2
To develop problem solving approach using analytical abilities, effective communication skills and teamwork.	3	2	2	3
To create awareness and understanding related to social issues, apart from developing a sense of commitment to the community and profession with sincere involvement.	3	3	3	2

Justification / Rationale for Mapping of Programme Educational Objectives (PEOs) with Departmental Missions

The mapping between the Programme Educational Objectives (PEOs) and the Departmental Missions of Mechanical Engineering has been carried out thoughtfully, considering the alignment of each PEO with the stated missions. The rationale for the assigned mapping levels (on a scale of 1 to 3) is detailed below:

PEO1:

Mission M1 strongly supports this PEO by imparting quality education and enhancing students' knowledge and skills, thereby preparing them to become globally competitive mechanical engineers. **Mission M2** contributes to this objective by providing access to modern research facilities and advanced learning resources that enrich students' academic experience and technical competence. **Mission M3** supports this P through interactions with industry, R&D organizations, and educational institutions, which expose students to current engineering practices and improve their employability. **Mission M4** complements this objec by providing a conducive learning environment that encourages academic excellence, creativity, and continuous self-improvement, enabling graduates to succeed in their professional careers.

PEO2:

Mission M1 lays the foundation for research by equipping students with strong theoretical and technical knowledge. **Mission M2** directly supports this PEO by maintaining state-of-the-art research facilities an encouraging the creation, interpretation, application, and dissemination of knowledge through research activities. **Mission M3** further strengthens this objective by promoting collaboration with industries, R&D organizations, and academic institutions, thereby facilitating applied research, innovation, consultancy, and knowledge exchange. **Mission M4** contributes by fostering creativity, critical thinking, and problem-solving abilities that are essential for conducting meaningful research and pursuing lifelong learning.

PEO3:

Mission M1 contributes to this PEO by providing a comprehensive engineering education that exposes students to modern technologies and contemporary engineering practices. **Mission M2** supports the obje by encouraging research activities that often involve multidisciplinary approaches and emerging technologies. **Mission M3** strongly aligns with this PEO through collaborations with industry and other institution providing opportunities to understand interdisciplinary applications and real-world engineering challenges. **Mission M4** further supports this objective by creating a learning environment that promotes awarene of societal, ethical, environmental, and technological issues, enabling students to develop a broader perspective on engineering practice.

PEO4:

Mission M1 supports this PEO by delivering quality education that develops analytical thinking, technical expertise, and communication skills among students. **Mission M2** contributes by engaging students in research and project activities that require systematic analysis and innovative problem-solving approaches. **Mission M3** strengthens this objective by facilitating industry interactions, collaborative projects, an consultancy activities that enhance teamwork, communication, and practical problem-solving abilities. **Mission M4** directly aligns with this PEO by fostering creativity, collaborative learning, and critical thinking which are essential for developing effective problem-solving skills and professional competence.

PEO5:

Mission M1 contributes to this PEO by promoting professional values and responsible engineering practices as part of the educational process. **Mission M2** indirectly supports this objective by encouraging research and innovation aimed at addressing societal and technological challenges. **Mission M3** enhances awareness of societal needs through interactions with industry, research organizations, and educationa institutions, helping students understand the role of engineers in community development and national progress. **Mission M4** strongly supports this PEO by providing an environment that nurtures ethical value social responsibility, leadership qualities, and a commitment to serving society and the engineering profession.

1.2 Curriculum Structure and Features (30)

1.2.1 State the Process for Developing/Revising the Program Curriculum (10)

The following systematic steps were adopted to review, revise, and align the B.Tech (Mechanical Engineering) curriculum with the desired Programme Outcomes (POs) and Programme Specific Outcomes (PSOs):

I. Feedbacks from course teachers, industry experts, and alumni working in the mechanical engineering domain were collected regarding course content, relevance, and modifications.

II. UGC/AICTE guidelines and National Education Policy (NEP-2020) were strictly followed during the syllabus revision. The detailed syllabus for each course was prepared by the concerned faculty members based on the objectives and expected course outcomes.

III. Lecture plans and delivery schedules were prepared after finalization of the revised syllabus.

IV. The Articulation Matrix was developed for each course, mapping individual Course Outcomes (COs) with Programme Outcomes (POs) and Programme Specific Outcomes (PSOs) on a scale of 0 to 3 [0 = No Relation, 2 = Medium/Moderate, 3 = High/Substantial].

V. Multiple departmental meetings were conducted to review the prepared documents and incorporate collective inputs from faculty members.

VI. A committee comprising 3–4 senior faculty members was constituted to critically review the draft curriculum. Gaps in the courses and syllabus with respect to attainment of POs and PSOs were identified and necessa were suggested and incorporated.

VII. All courses were compiled as per the new programme structure. The Course Articulation Matrices of individual courses were consolidated into the Programme Articulation Matrix.

VIII. Any identified gaps in compliance were documented. Measures to bridge these gaps were incorporated through additional activities and clearly mentioned in the Course Information Sheet. Examples of such brid include: Mini projects / Major projects, Industry expert / Guest lectures, Advanced laboratory experiments, Industrial visits, Summer / Internship training, Self-learning modules (NPTEL, MIT OpenCourseWare, SWA Simulations / Case studies.

IX. The draft syllabus was thoroughly discussed and approved in the Departmental Meeting.

X. The programme structure along with the complete syllabus was presented and approved in the meeting of the Board of Studies (BOS) of Mechanical Engineering.

XI. The approved B.Tech (Mechanical Engineering) syllabus and course structure were placed before the Academic Council of BIT Mesra for approval. Subsequently, the final approved syllabus along with the course struc before the Board of Governors (BOG) of BIT Mesra for final ratification.

1.2.2 Curriculum Structure (10)

Edit

ID	Course Code	Course Title	Classroom Instruction (CI) (in hours per semester)		Lab Instruction (LI) (in hours per semester)	Term Work (TW) and Self Learning (SL) (TW+ SL) (in hours per semester)	Total no. of Hou semester
			L	T	P	SL	
1	MA24101	Mathematics-I	42	14	0	64	120
2	PH24101	Physics	42	14	0	64	120
3	BE24101	Biological Science for Engineers	28	0	0	32	60
4	CS24101	Programming for Problem Solving	42	14	0	64	120
5	EE24101	Basics of Electrical Engineering	28	14	0	48	90
6	PH24102	Physics Laboratory	0	0	28	2	30
7	CS24102	Programming for Problem Solving Lab	0	0	28	2	30
8	HS24131	Communication Skills-I	0	0	42	3	45
9	MC 24 101/102/ 103/104/105	NCC/NSS/PT & Games/Creative Arts	0	0	28	2	30
10	MA24103	Mathematics-II	42	14	0	64	120
11	CH24101	Chemistry	42	14	0	64	120
12	CE24101	Environmental Science	28	0	0	32	60
13	ME24101	Basics of Mechanical Engineering	28	14	0	48	90
14	EC24101	Basics of Electronics and Communication Engineering	28	14	0	48	90
15	CH24102	Chemistry Laboratory	0	0	28	2	30
16	EC24102	Electronics and Communication Lab	0	0	28	2	30
17	ME24102	Engineering Graphics	0	0	56	4	60
18	MC 24 105/106/ 107/108	NCC/NSS/PT & Games/Creative Arts	0	0	28	2	30
19	ME24201	Thermodynamics	42	0	0	48	90
20	ME24203	Fluid Mechanics and Hydraulic Machines	42	0	0	48	90
21	ME24205	Strength of Materials	42	14	0	64	120

22	PE24201	Introduction to Materials Engineering	42	0	0	48	90
23	PE22297	Manufacturing Processes	42	0	0	48	90
24	MA24201	Numerical Methods	28	0	0	32	60
25	MT24131	UHV-II: Understanding Harmony	42	0	0	48	90
26	MA24202	Numerical Methods Lab	0	0	28	2	30
27	ME24202	Fluid Mechanics and Hydraulic Machines Lab	0	0	28	2	30
28	MC 24 201/202/ 203/204	NCC/NSS/PT & Games/Creative Arts	0	0	28	2	30
29	ME24207	Kinematics and Dynamics of Machines	42	14	0	64	120
30	ME24209	Thermal Energy Conversion Systems	42	14	0	64	120
31	ME24211	Machine Design	42	14	0	64	120
32	HS24211	Indian Knowledge System	0	0	0	0	0
33	ME24,215, 217, 219	Program Elective-I	42	14	0	64	120
34	XXXXXX /MO24201	Open Elective-I / MOOC-I	42	0	0	48	90
35	ME24210	Dynamics of Machines Lab	0	0	42	3	45
36	PE24298	Manufacturing Processes Lab	0	0	42	3	45
37	MC 24 206 / 207/208/209/210	NCC/NSS/PT & Games/Creative Arts	0	0	28	2	30
38	ME24301	IC Engines and Gas Turbines	42	14	0	64	120
39	ME24303	Mechanical Vibration	42	0	0	48	90
40	ME24305	Heat and Mass Transfer	42	0	0	48	90
41	ME24 313, 315, 317, 319,	Program Elective-II	42	0	0	48	90
42	ME24 PE24313 321,323, 325	Program Elective-III	42	0	0	48	90
43	XXXXXXX /MO24301	Open Elective-II / MOOC-II	42	0	0	48	90
44	ME24302	Heat Transfer Lab	0	0	28	2	30
45	ME24304	Internal Combustion Engines Lab	0	0	28	2	30
46	HS24133	Communication Skill - II	0	0	42	3	45

47	ME24300	Project - I	0	0	0	60	60
48	ME24307	Computer Aided Design	28	0	0	32	60
49	ME24309	Automobile Engineering	42	14	0	64	120
50	ME24311	Robotics Engineering	42	14	0	64	120
51	ME24, 331, 333, 329, PE24335	Program Elective-IV	42	0	0	48	90
52	ME24 335, 337,339 PE24345	Program Elective-V	42	0	0	48	90
53	XX24XXX/ MO24303	Open Elective-III / MOOC-III	42	0	0	48	90
54	ME24306	Robotics and Automation Lab	0	0	42	3	45
55	ME24308	Automobile Engineering Lab	0	0	42	3	45
56	ME24350	Project - II	0	0	0	60	60
57	ME24310	Computer Aided Design & Drafting Lab	0	0	28	2	30
58	ME24401	Refrigeration and Air conditioning	42	14	0	64	120
59	ME24 403, 405, 407, PE24 407 413	Program Elective-VI	42	0	0	48	90
60	XXXXX/MO24401	Open Elective-IV / MOOC-IV	42	0	0	48	90
61	ME24404	Refrigeration and air conditioning Lab	0	0	28	2	30
62	ME24400	Project - III	0	0	0	90	90
63	ME24400	Summer Training (Minimum Four Weeks / 160 Hrs)	0	0	0	120	120
64	MT24204	Constitution of india	0	0	0	0	0
65	EE24102	Electrical Engineering Laboratory	0	0	28	2	30
66	PE24102	Workshop Practice	0	0	28	2	30
67	ME24208	Basic Mechanical Engineering lab	0	0	28	2	30
68	ME24450/ ME24490	Project-IV / Industry Internship	0	0	0	180	180
69	ME24498	Comprehensive Viva	0	0	0	60	60
		Total	**1456**	**238**	**784**	**2562**	**5040**

1.2.3 Components of Curriculum (5)

Course Components	Curriculum Content (% of total number of credits of the program)	Total number of contact hours	Total numbe
Basic Sciences	14.88	750.00	
Basic Engineering	11.31	570.00	
Humanities and Social Scie	3.57	180.00	
Program Core	38.69	1950.00	
Program Electives	13.1	660.00	
Open Electives	7.14	360.00	
Project(s)	7.73	390.00	
Internships/Seminars	2.38	120.00	
Any other (Please specify)	1.19	60.00	
Total number of Credits			

1.2.4 Strategies for Education Reforms (5)

The Department of Mechanical Engineering has implemented several key educational reforms in alignment with the National Education Policy (NEP 2020) to enhance learning flexibility, industry relevance, and multidisciplinary exposure. The following strategies have been adopted:

1.

Modification of Course Curriculum:

The B.Tech (Mechanical Engineering) curriculum has been comprehensively redesigned as per NEP 2020 guidelines. The new structure offers greater flexibility to students through a wide range of elective courses, open elect choice-based credit systems. Provisions for Minor degrees, Honours with in-depth specialization, and vocational/skill-oriented courses have been incorporated. Multiple entry and exit options have been introduced, allowing s exit after completing the required credits with appropriate certifications: a Certificate after 1 year, a Diploma after 2 years, and a B.Sc. Engineering degree after 3 years. The curriculum strongly promotes interdisciplinary lea integrating foundation courses such as Communication Skills, Indian Knowledge System, Universal Human Values (UHV), Ethics, and discipline-specific core courses.

2.

Provision of Project-Based Learning and Industrial Internship:

The curriculum emphasizes hands-on, experiential learning through project-based learning that encourages students to integrate knowledge from multiple disciplines to solve real-world mechanical engineering problems. A r Summer Industrial Internship of 2 credits has been introduced. Additionally, students have the option to undertake their final year major project (6 credits) in the form of industry-sponsored or internship-based projects, the bridging the gap between academia and industry.

3.

Academic Bank of Credits (ABC) and APAAR:

In line with NEP 2020, the Academic Bank of Credits (ABC) has been implemented, enabling seamless credit transfer between recognized institutions and facilitating multiple entry–exit options.All credits earned by students stored in the ABC, providing a transparent and flexible credit accumulation system.Each faculty is assigned a unique APAAR ID (Automated Permanent Academic Account Registry), which serves as a lifelong academic identit tracking and managing their academic journey across institutions.

4.

Expected Outcomes:

Greater flexibility and personalized learning pathways for Mechanical Engineering students.
Promotion of multidisciplinary and interdisciplinary education in line with NEP 2020.
Enhanced student mobility through seamless credit transfer across institutions. Improved industry readiness and practical skills through project-based learning and mandatory internships.

1.3 PO, PSO and their Mapping with Courses (20)

1.3.1 POs and PSOs (5)

PSO1	

1.3.2 Mapping between the Courses and POs/PSOs (15)

PO Number	List of Courses
PO1	ME24101, ME24102, ME24104, ME24201, ME24203, ME24205, ME24202, ME24207, ME24209, ME24211, ME24215, ME24217, ME24219, ME24212, ME24208, ME24210, ME24301, ME24303, ME24305, ME24313, M ME24317, ME24319, ME24321, ME24323, ME24325, ME24341, ME24343, ME24302, ME24304, ME24300, ME24307, ME24309, ME24311, ME24331, ME24333, ME24329, ME24335, ME24337, ME24339, ME24306, M ME24310, ME24350, ME24401, ME24403, ME24405, ME24407, ME24404, ME24400, ME24450, ME24490, ME24498
PO2	ME24101, ME24102, ME24104, ME24201, ME24203, ME24205, ME24202, ME24207, ME24209, ME24211, ME24215, ME24217, ME24219, ME24212, ME24208, ME24210, ME24301, ME24303, ME24305, ME24313, M ME24317, ME24319, ME24321, ME24323, ME24325, ME24341, ME24343, ME24302, ME24304, ME24300, ME24307, ME24309, ME24311, ME24331, ME24333, ME24329, ME24335, ME24337, ME24339, ME24306, M ME24310, ME24350, ME24401, ME24403, ME24405, ME24407, ME24404, ME24400, ME24450, ME24490
PO3	ME24101, ME24102, ME24201, ME24203, ME24205, ME24207, ME24209, ME24211, ME24215, ME24217, ME24219, ME24301, ME24303, ME24305, ME24313, ME24315, ME24317, ME24319, ME24321, ME24323, M ME24341, ME24343, ME24307, ME24309, ME24311, ME24331, ME24333, ME24329, ME24335, ME24337, ME24339, ME24401, ME24403, ME24405, ME24407, ME24300, ME24350, ME24400, ME24450
PO4	ME24202, ME24208, ME24210, ME24302, ME24304, ME24306, ME24308, ME24310, ME24404, ME24300, ME24350, ME24400, ME24450, ME24490
PO5	ME24102, ME24104, ME24212, ME24307, ME24310, ME24311, ME24329, ME24300, ME24350, ME24400, ME24450
PO6	ME24101, ME24301, ME24309, ME24401, ME24403, ME24407, ME24490
PO7	ME24300, ME24350, ME24400, ME24450, ME24490
PO8	ME24102, ME24104, ME24202, ME24208, ME24210, ME24302, ME24304, ME24306, ME24308, ME24310, ME24404, ME24300, ME24350, ME24400, ME24450, ME24490
PO9	ME24102, ME24104, ME24202, ME24208, ME24210, ME24302, ME24304, ME24306, ME24308, ME24310, ME24404, ME24300, ME24350, ME24400, ME24450, ME24490, ME24498
PO10	ME24300, ME24350, ME24400, ME24450, ME24490
PO11	ME24101, ME24102, ME24104, ME24201, ME24203, ME24205, ME24202, ME24207, ME24209, ME24211, ME24215, ME24217, ME24219, ME24212, ME24208, ME24210, ME24301, ME24303, ME24305, ME24313, M ME24317, ME24319, ME24321, ME24323, ME24325, ME24341, ME24343, ME24302, ME24304, ME24300, ME24307, ME24309, ME24311, ME24331, ME24333, ME24329, ME24335, ME24337, ME24339, ME24306, M ME24310, ME24350, ME24401, ME24403, ME24405, ME24407, ME24404, ME24400, ME24450, ME24490, ME24498

PO Number	List of Courses
PSO1	ME24101, ME24102, ME24104, ME24201, ME24202, ME24203, ME24205, ME24207, ME24208, ME24209, ME24210, ME24211, ME24212, ME24215, ME24217, ME24219, ME24223, ME24225, ME24300, ME24301, M ME24303, ME24304, ME24305, ME24306, ME24307, ME24308, ME24309, ME24310, ME24311, ME24313, ME24315, ME24317, ME24319, ME24321, ME24323, ME24325, ME24329, ME24331, ME24333, ME24335, M ME24339, ME24341, ME24343, ME24345, ME24350, ME24400, ME24401, ME24403, ME24404, ME24405, ME24407, ME24411, ME24413, ME24450, ME24490, ME24498
PSO2	ME24101, ME24102, ME24104, ME24201, ME24202, ME24203, ME24205, ME24207, ME24208, ME24209, ME24210, ME24211, ME24212, ME24215, ME24217, ME24219, ME24223, ME24225, ME24300, ME24301, M ME24303, ME24304, ME24305, ME24306, ME24307, ME24308, ME24309, ME24310, ME24311, ME24313, ME24315, ME24317, ME24319, ME24321, ME24323, ME24325, ME24329, ME24331, ME24333, ME24335, M ME24339, ME24341, ME24343, ME24345, ME24350, ME24400, ME24401, ME24403, ME24404, ME24405, ME24407, ME24411, ME24413, ME24450, ME24490, ME24498
PSO3	ME24101, ME24102, ME24104, ME24201, ME24202, ME24203, ME24205, ME24207, ME24208, ME24209, ME24210, ME24211, ME24212, ME24215, ME24217, ME24219, ME24223, ME24225, ME24300, ME24301, M ME24303, ME24304, ME24305, ME24306, ME24307, ME24308, ME24309, ME24310, ME24311, ME24313, ME24315, ME24317, ME24319, ME24321, ME24323, ME24325, ME24329, ME24331, ME24333, ME24335, M ME24339, ME24341, ME24343, ME24345, ME24350, ME24400, ME24401, ME24403, ME24404, ME24405, ME24407, ME24411, ME24413, ME24450, ME24490, ME24498

1.4 Course Outcomes and Course Articulation Matrix (30)

1.4.1 Course Outcome (Semester Wise) (15)

No. of Core Courses : 9	**C2 :** 4	**C3 :** 4	**C4 :** 1

Note : Number of Outcomes for a Course is expected to be around 6.

Course Code :	ME24201	Semester :	3

Course Outcome	Statements
ME24201.1	Discuss the basic concepts of thermodynamics.
ME24201.2	Apply the laws of thermodynamics to various flow, non-flow and cyclic systems.
ME24201.3	Evaluate the performance different engineering devices.
ME24201.4	Develop thermodynamics relations for engineering practices.

Course Code :	ME24203	Semester :	3

Course Outcome	Statements
ME24203.1	Discuss the fundamental properties and behavior of fluids in static and dynamic conditions.
ME24203.2	Apply the principles of fluid kinematics and dynamics to solve engineering problems involving fluid flows.
ME24203.3	Analyse the problems related to fluid flow devices.
ME24203.4	Develop mathematical models and relations for fluid flow and hydraulic systems applicable in engineering practices.

Course Code :	ME24207	Semester :	4

Course Outcome	Statements
ME24207.1	Explain kinematic and dynamic principles of planar mechanism.
ME24207.2	Apply the basic principles into planar mechanisms for relevant applications.
ME24207.3	Anlayse the acceleration and forces during the motion of machines and mechanisms.
ME24207.4	Evaluate the problems related to the motions of machines and mechanism.

Course Code :	ME24211	Semester :	4

Course Outcome	Statements
ME24211.1	Explain the principles of machine design and failure theories.
ME24211.2	Apply the design concepts on various joints and components of machines.
ME24211.3	Analyse the forces and strength of for different parts of the machine components.
ME24211.4	Evaluate different conditions problems associated to machine components.
ME24211.5	Create or design the components according to specific/special conditions.

Course Code :	ME24301	Semester :	5

Course Outcome	Statements
ME24301.1	Explain the working principle of various types of IC Engine and gas turbines.
ME24301.2	Apply the fundamental principles of thermodynamics, combustion and fluid flow in different engine and gas turbine processes.
ME24301.3	Examine the key factors involved in enhancing performance of engine and gas turbine.
ME24301.4	Solve numerical problems based on engine and gas turbine testing and performance.
ME24301.5	Explain design features and performance characteristics of SI and CI engines.

Course Code :	ME24303	Semester :	5

Course Outcome	Statements
ME24303.1	Explain the basic elements of vibration models.
ME24303.2	Analyze multi degrees of freedom systems.
ME24303.3	Solve the natural frequencies and mode shapes of a vibrating system.
ME24303.4	Analyse the vibration of continuous systems.
ME24303.5	Apply the knowledge of vibration in practical problems.

Course Code :	ME24309	Semester :	6

Course Outcome	Statements
ME24309.1	Understand the principles and applications of automobile engineering, including vehicle systems, components, and their interactions.
ME24309.2	Apply mechanical engineering principles to solve problems related to vehicle dynamics.
ME24309.3	Apply electrical and electronic engineering concepts to automotive systems.
ME24309.4	Solve complex issues related to Automobile, such as design challenges and improving system reliability.
ME24309.5	Investigate and integrate emerging technologies in automotive engineering

Course Code :	ME24311	Semester :	6

Course Outcome	Statements
ME24311.1	Explain the fundamental concepts and mathematical foundations of Robotics Engineering, including degrees of freedom, transformation matrices, homogeneous coordinates, and Euler angle representation.
ME24311.2	Utilize the Denavit-Hartenberg representation and various kinematic approaches to solve problems in forward and inverse kinematics of serial robots.

ME24311.3	Evaluate the role of the Jacobian matrix in robot statics, velocity, and acceleration analysis, and solve gravity compensation problems in robotic systems.
ME24311.4	Derive equations of motion for serial robots using Lagrange-Euler and Newton-Euler formulations, and assess different control methods for trajectory following in manipulators.
ME24311.5	Design and implement control strategies for robotic systems, including feedback and closed-loop control, and integrate robotic systems into industrial applications such as PUMA and SCARA configurations.

Course Code :	**ME24401**	**Semester :**	**7**

Course Outcome	**Statements**
ME24401.1	Describe the fundamental principles of refrigeration and Air-conditioning, including the thermodynamic cycles.
ME24401.2	Apply the basic concepts in designing the refrigeration and air- conditioning systems for various applications, including commercial, industrial, and domestic use.
ME24401.3	Analyse the energy consumption of refrigeration and air conditioning systems and evaluate methods to improve energy efficiency.
ME24401.4	Solve complex issues related to refrigeration and air conditioning systems, such as optimizing performance, solving design challenges, and improving system reliability.

1.4.2 Course Articulation Matrix (15)

1 . course name : C2ME24201

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
C2ME24201.1	Discuss the	3	2	-	-	-	-	1	1	-	-	2
C2ME24201.2	Apply the la	3	2	-	-	-	-	1	1	-	-	2
C2ME24201.3	Evaluate th	3	2	1	1	-	-	1	1	-	-	2
C2ME24201.4	Develop the	3	2	-	-	-	-	1	1	-	-	2
Average		**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**

2 . course name : C2ME24203

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
C2ME24203.1	Discuss the	3	2	-	2	-	-	-	-	-	-	2
C2ME24203.2	Apply the p	3	3	2	2	2	-	-	1	2	-	3
C2ME24203.3	Analyse the	2	3	3	2	2	-	-	2	2	2	2
C2ME24203.4	Develop ma	3	3	3	3	3	-	-	1	2	2	3
Average		**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**

3 . course name : C2ME24207

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
C2ME24207.1	Explain kine	3	3	3	3	2	2	2	2	2	1	2
C2ME24207.2	Apply the b	3	3	3	3	2	3	2	2	2	1	2
C2ME24207.3	Anlayse the	3	3	3	3	2	2	2	2	2	1	2
C2ME24207.4	Evaluate th	3	3	3	3	2	2	2	2	2	1	2
Average		**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**

4 . course name : C2ME24211

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
C2ME24211.1	Explain the	3	2	2	1	2	1	1	1	1	1	2
C2ME24211.2	Apply the d	3	2	3	2	3	1	1	2	1	2	2
C2ME24211.3	Analyse the	3	3	2	2	2	1	1	1	1	2	2
C2ME24211.4	Evaluate di	3	2	3	2	2	2	1	1	1	2	2
C2ME24211.5	Create or d	3	2	3	2	3	2	1	2	2	2	3

Average		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

5 . course name : C3ME24301

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
C3ME24301.1	Explain the	3	3	2	1	1	1	-	1	-	-	3
C3ME24301.2	Apply the fu	3	3	2	2	1	1	-	1	-	-	3
C3ME24301.3	Examine th	3	3	2	2	1	1	-	1	-	-	2
C3ME24301.4	Solve nume	3	3	3	3	1	1	1	1	2	1	3
C3ME24301.5	Explain des	3	3	3	3	1	1	1	1	2	1	2
Average		**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**

6 . course name : C3ME24303

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
C3ME24303.1	Explain the	3	2	1	2	1	-	-	1	1	-	2
C3ME24303.2	Analyze mu	3	3	2	2	3	1	1	2	2	1	2
C3ME24303.3	Solve the n	3	3	3	3	3	2	1	2	3	2	3
C3ME24303.4	Analyse the	3	2	3	3	2	3	2	3	3	3	3
C3ME24303.5	Apply the k	3	2	3	3	2	3	2	3	3	3	3
Average		**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**

7 . course name : C3ME24309

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
C3ME24309.1	Understanc	3	3	3	3	3	2	2	2	3	3	3
C3ME24309.2	Apply mech	3	3	3	3	3	3	2	3	3	3	3
C3ME24309.3	Apply electi	3	3	3	3	3	2	2	3	3	3	3
C3ME24309.4	Solve comp	3	3	3	3	3	3	2	3	3	3	3
C3ME24309.5	Investigate	3	3	3	3	3	3	2	3	3	3	3
Average		**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**

8 . course name : C3ME24311

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
C3ME24311.1	Explain the	3	2	1	1	2	1	1	1	2	1	3
C3ME24311.2	Utilize the D	2	3	3	2	3	2	1	2	3	2	3
C3ME24311.3	Evaluate th	2	3	2	3	2	3	1	1	2	2	3
C3ME24311.4	Derive equa	2	3	3	3	2	1	1	2	3	3	3
C3ME24311.5	Design and	3	3	3	3	3	2	2	2	3	3	3
Average		**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**

9 . course name : C4ME24401

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
C4ME24401.1	Describe th	3	2	1	1	2	1	-	-	-	-	2
C4ME24401.2	Apply the b	3	3	3	2	2	2	-	1	1	2	2
C4ME24401.3	Analyse the	2	3	2	3	2	3	-	-	-	1	3
C4ME24401.4	Solve comp	3	3	3	3	3	2	1	1	1	2	3
Average		**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**

1 . Course Name : C2ME24201

Course
C2ME24201.1
C2ME24201.2
C2ME24201.3
C2ME24201.4

2 . Course Name : C2ME24203

Course
C2ME24203.1
C2ME24203.2
C2ME24203.3
C2ME24203.4

3 . Course Name : C2ME24207

Course
C2ME24207.1
C2ME24207.2
C2ME24207.3
C2ME24207.4

4 . Course Name : C2ME24211

Course
C2ME24211.1
C2ME24211.2
C2ME24211.3
C2ME24211.4
C2ME24211.5

5 . Course Name : C3ME24301

Course
C3ME24301.1
C3ME24301.2
C3ME24301.3
C3ME24301.4
C3ME24301.5

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6 . Course Name : C3ME24303

Course
C3ME24303.1
C3ME24303.2
C3ME24303.3
C3ME24303.4
C3ME24303.5

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7 . Course Name : C3ME24309

Course
C3ME24309.1
C3ME24309.2
C3ME24309.3
C3ME24309.4
C3ME24309.5

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8 . Course Name : C3ME24311

Course
C3ME24311.1
C3ME24311.2
C3ME24311.3
C3ME24311.4
C3ME24311.5

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9 . Course Name : C4ME24401

Course
C4ME24401.1
C4ME24401.2
C4ME24401.3
C4ME24401.4

1.5 Program Articulation Matrix (5)

Program Articulation Matrix

Course Code	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
ME24201	3	2	1	1	PO5	PO6	1	1	PO9	PO10
ME24203	2.75	2.75	2.67	2.25	2.33	PO6	PO7	1.33	2	2
ME24205	3	2.5	2	2.25	2	PO6	PO7	PO8	PO9	PO10
PE24201	3	2.8	2.2	3	1.75	1.75	1	2.25	1.67	2
ME24297	2.6	2.6	2.2	1.4	2.2	1	1	1	1.4	1.6
MA24201	3	2.6	2	2.2	2.6	PO6	PO7	PO8	1	1
MT24131	PO1	PO2	2	1	1	3	3	2	2	1
MA24202	3	2.4	1.8	2.2	3	PO6	PO7	1	1	1.2
ME24202	3	3	3	3	3	2.25	2.25	2.25	2.25	1.25
ME24207	3	3	3	3	2	2.25	2	2	2	1
ME24209	3	2	2.25	1.25	PO5	PO6	PO7	1	PO9	PO10
ME24211	3	2.2	2.6	1.8	2.4	1.4	1	1.4	1.2	1.8
ME24215	3	2.75	2.25	1.75	1.5	2.25	1.5	1.75	2.25	1.25
ME24217	3	2.5	2	1.5	2	2.75	1	1	1.25	1.75
ME24219	3	2.75	2.5	2	2	2	PO7	PO8	2	PO10
ME24208	3	3	3	3	3	2.25	2.25	2.25	2.25	1.25
ME24210	3	3	3	3	3	2.4	2.4	2.4	2.4	1.4
PE24298	2.6	2.4	2.8	2.4	2.6	1	1	1.6	1.6	1.6
ME24301	3	3	2.4	2.2	1	1	1	1	2	1
ME24303	3	2.4	2.4	2.6	2.2	2.25	1.5	2.2	2.4	2.25
ME24305	3	3	3	1.75	1.25	1	1	2	1.75	1.5
ME24313	2.75	2	2	2.5	2.5	1.67	PO7	PO8	2	PO10
ME24315	3	3	1.6	2.6	2.33	1.5	1	1	1	1
ME24317	3	2.75	2.75	2.75	2.75	1	PO7	PO8	1	PO10
ME24319	3	3	2	2	1.6	PO6	1	1	PO9	PO10
PE24313	2.8	2.2	2.4	1.5	2.4	1.8	1	1.33	1	1.33
ME24321	3	3	3	3	2.2	PO6	PO7	PO8	PO9	PO10
ME24323	3	2.75	2.75	2.5	3	PO6	PO7	PO8	2.25	PO10
ME24325	3	2.75	2.25	1.75	1.75	1.25	PO7	PO8	PO9	PO10
PE24315	2.2	2.2	2.4	2	2.4	1	1	PO8	1	1
ME24302	3	3	3	3	3	2.25	2.25	2.25	2.25	1.25

ME24304	2.25	1.5	1.75	1.25	1.75	1.75	1.75	2	1.5	1.5
ME24307	2.25	2.25	2.25	2	2.67	1.25	1	PO8	1	PO10
ME24309	3	3	3	3	3	2.6	2	2.8	3	3
ME24311	2.4	2.8	2.4	2.4	2.4	1.8	1.2	1.6	2.6	2.2
ME24329	3	3	3	3	1.8	PO6	PO7	PO8	1.25	PO10
ME24331	2.75	2.75	2.75	2.5	1.75	2.5	2	2.5	2.5	3
ME24333	3	3	3	3	2	2	PO7	2	3	PO10
PE24335	3	3	3	1	2	3	1	2	2	3
ME24335	3	2.75	2.5	2.5	2.75	1.5	1	1.75	2.25	1.25
ME24337	3	3	3	3	1.8	PO6	PO7	PO8	1.25	PO10
ME24339	3	3	3	3	2	2	PO7	2	3	PO10
PE24345	3	2.6	2.2	1.5	3	1	1	2	PO9	2.6
ME24306	2.5	2.5	2.5	2	2.5	1	1	2.25	2.25	1.75
ME24308	3	3	3	3	3	2.5	2	2.75	3	3
ME24310	3	2.8	2.8	2.8	3	2	PO7	2	1.2	PO10
ME24401	2.75	2.75	2.25	2.25	2.25	2	1	1	1	1.67
ME24403	2.6	2.4	2.6	2	2.33	2.2	1.67	1.2	1.6	2.2
ME24405	3	2.75	2.25	2.5	2.75	1.75	1	1	1	1.5
ME24407	3	2.75	2.25	2.25	2	1.5	1	PO8	1	1
PE24407	3	3	3	2.2	1	PO6	PO7	PO8	PO9	3
MT24204	2	2	1	2	1.4	2	2	2	1.8	1.6
ME24404	3	2.6	2.2	2.4	2.8	2	1	1	1	1.67
ME24450	2	2.4	2.2	2.2	2.2	1.6	1.6	1.6	1.8	2

Course Code	PSO1	PSO2	PSO3
MA24201	2.4	1.4	2.2
MA24202	3	2.4	2.6
ME24201	2.75	2	2.5
ME24202	3	2.25	3
ME24203	2.75	2	2.5
ME24205	2.5	1.25	2
ME24207	2	2	3
ME24208	3	2.25	3

ME24209	3	2.25	2
ME24210	3	2.4	3
ME24211	3	2.6	2.2
ME24215	3	1	2
ME24217	2.75	2.5	2.25
ME24219	3	2.25	2
ME24297	3	2.6	2.2
ME24301	2	1	1
ME24302	3	2.25	3
ME24303	3	2.8	2.4
ME24304	2	2	2
ME24305	3	2	2
ME24306	2.75	2.75	2.75
ME24307	2.5	1.75	2
ME24308	3	3	3
ME24309	3	3	3
ME24310	2.6	2.6	3
ME24311	2.6	2.6	2.6
ME24313	2.5	1.75	2
ME24315	3	2	2
ME24317	3	2.75	3
ME24319	3	2.6	3
ME24321	3	1.4	3
ME24323	3	2.25	2
ME24325	3	2.25	2
ME24329	3	1.8	1.8
ME24331	3	3	3
ME24333	3	3	3
ME24335	3	2.5	2.5
ME24337	3	1.8	1.8
ME24339	3	3	3
ME24401	2.75	2.75	3

ME24403	2	3	3
ME24404	3	2.4	3
ME24405	3	2.5	3
ME24407	3	2.75	3
ME24450	2.4	2.4	2.8
MT24131	1	2	2
MT24204	1	1.6	2.4
PE24201	2.6	2.4	2.8
PE24298	3	2.2	1.6
PE24313	3	2.2	2
PE24315	3	2	2
PE24335	3	2	2
PE24345	3	2	2.8
PE24407	3	2.2	2

2 OUTCOME-BASED TEACHING LEARNING (120)

2.1 Describe Processes Followed to Ensure Quality of Teaching & Learning (20)